

Abhishek Deo

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Hands-on Principal Architect · 19+ yrs · Problem-first production AI: RAG, multi-agent, LLM · Java + Spring Boot (11 yrs) · Multi-tenant SaaS · AWS (ECS Fargate), Terraform · Graph + GNN · HIPAA, ISO 27001 · AI-Leveraged Delivery (Claude Code + OpenAI Codex)

SUMMARY

Hands-on principal architect, **19+ years**, who starts from the business problem and designs back to the architecture. Strongest in **Java and Spring Boot**: about **11 years of Java across three eras**, from server-side Java in 2008 to **Spring Boot 2.x on Java 8/17** through February 2025. At Deque, every customer ran its own instance, so each upgrade and onboarding was repeated per customer; consolidated per-customer deployments onto shared **multi-tenant** software as a service (SaaS) infrastructure, with dedicated isolation for regulated banking customers. At VoltusWave, a chronic-care client needed artificial intelligence (AI) in its product to raise investment, which first needed patients on a reliable chat platform: rewrote its core chat service from **Node.js to Go** behind a two-stream **Amazon Kinesis** design, multi-availability-zone with failover and disaster recovery for **Health Insurance Portability and Accountability Act (HIPAA)** compliance. Then built ten analytics and AI services on Amazon Web Services (AWS), all in production: retrieval-augmented generation (RAG) precedent search for clinicians, a multi-agent orchestrator, causal inference and leakage-safe machine learning over **9–10 years of patient history**, and self-hosted **large language models (LLMs)** inside the client boundary, where code computes every number and a model only narrates; alongside them, a **graph neural network (GNN)** context graph. Standardised the organisation on **Terraform and Amazon Elastic Container Service (ECS) Fargate**, cutting infrastructure cost **25–30%** and deploy time to **under 10 minutes**.

KEY SKILLS

Technology Leadership & Strategy

- **Technology Strategy & Architecture**
- **Decision-Intelligence Platforms** (Event-Sourced + Graph + AI; Retrieval, Cohort Aggregation, Sequence Mining and Graph Traversal archetypes)
- Cross-Team Technical Leadership (**6 sub-teams**, company-wide standards)
- Engineering Mentorship & Team Scaling (**0 to 35 engineers**, 2017–18)

Engineering Excellence

- **Multi-Tenant SaaS Architecture** (Shared-Schema, Per-Tenant Isolation, Regulated Carve-Outs, Shared or Dedicated Amazon Neptune per Tenant)
- **Java + Spring Boot Microservices** (Java 8/17, Spring Boot 2.x, JavaServer Pages, Versioned Application Programming Interfaces (APIs); microservices since 2016, representational state transfer (REST) since 2013)
- **Distributed Systems at Scale** (High Availability, Multi-Availability-Zone Failover, Disaster Recovery, Load Balancing)
- **Go, Node.js + TypeScript Services** (Amazon Kinesis Enhanced Fan-Out, Lambda, Simple Queue Service)
- **Terraform, AWS CloudFormation + Docker on AWS** (company-wide standard, Elastic Container Service (ECS) Fargate)
- **Cloud Data Stores** (PostgreSQL, Amazon Aurora, DynamoDB, Amazon OpenSearch, Redis, Amazon Neptune + Gremlin, DuckDB + Parquet)
- **Event Sourcing + Command Query Responsibility Segregation** (CQRS, DynamoDB Single-Table, Parquet Snapshot Exporters)
- **AI/Machine Learning in Product** (Graph Neural Networks on Amazon Neptune, Clinical Decision Intelligence)
- **Retrieval-Augmented Generation & Vector Search** (Elasticsearch dense_vector, Amazon SageMaker Embeddings, warm-path P99 target under 200 ms)
- **Self-Hosted Models In-Virtual Private Cloud** (Ollama, Llama 3.1, Self-Hosted LangSmith)
- **Python AI Services** (FastAPI, Django, LangChain, LangGraph, Model Context Protocol Servers)
- **Causal Inference & Leakage-Safe Machine Learning** (Propensity Matching, As-Of Featurisation, Outcome-Maturity Windows)
- **LLM Productionisation** (Guardrails, Field-Level Claim Grammar, Replay Determinism)
- **Identity & Access** (Keycloak to Red Hat single sign-on, OpenID Connect, OAuth 2.0)
- **Continuous Integration / Continuous Delivery (CI/CD) & DevOps** (Jenkins to GitHub Actions, Puppeteer Regression Automation)
- **Observability & Reliability Engineering** (Datadog, Amplitude, ELK: Elasticsearch, Logstash, Kibana)
- **AI-Leveraged Delivery** (Claude Code + OpenAI Codex across a **20-engineer** team)

Business & Delivery

- **Security, Compliance & Data Privacy** (HIPAA controls, ISO 27001 release pipeline, least-privilege, role-based access control)
- **Enterprise Onboarding & Consolidation** (**280+ axe Monitor Clients**, Every-Instance Migration, Internal Environments)
- **Accessibility Compliance** (Web Content Accessibility Guidelines 2.2 Level AA)
- **Agile / Scrum Delivery** across distributed United States–India teams

PROFESSIONAL EXPERIENCE

Vice President of Technology

Mar '25 - Apr '26

VoltusWave Technologies

Hyderabad

- Distilled the client's goal of raising investment on artificial intelligence (AI) into **10 killer queries**, agreed with its chief executive, proving the AI was real
- Pioneered a **Context Graph on Amazon Neptune** with graph neural network (GNN) models across **80+ chronic conditions**, projected from an event-sourced DynamoDB single-table spine
- Delivered **10 decision-intelligence services** in Python and FastAPI on 4 query archetypes to production for ~10,000 patients, absorbing 20 review findings into one design
- Engineered retrieval-augmented generation (RAG) for clinical precedent search across **80+ chronic conditions**, because a database filter returns a cohort, not the most analogous decisions
- Optimised retrieval around the decision, not the patient, via tenant-filtered **Elasticsearch dense_vector** nearest-neighbour search, tuned to a 0.75 normalised discounted cumulative gain target

- Launched the precedent panel **in production** for doctors, health coaches and guidance counsellors, answering cases short of 7 required fields with a plain cohort count
- Conceived a **multi-agent orchestrator** showing which care decisions followed when a patient's payments, adherence and support broke inside one 90-day window
- Developed orchestration as a deterministic LangGraph state graph: **3 blind finder agents** run in parallel; a barrier re-verifies every proposed patient against a frozen snapshot
- Shipped the orchestrator to production behind **executive and operations dashboards** and audited per-patient drill-ins, suppressing any delta failing a 0.10 balance gate
- Secured protected health information inside the client virtual private cloud, self-hosting **Llama 3.1** on Ollama, embeddings and LangSmith tracing so patient data never left
- Restricted large language models (LLMs) in production to narrating code-computed numbers — pinned model, **temperature 0**, chain hash, field-level claim grammar rejecting uncited claims
- Instituted confounder-gated cohort comparisons, since clinicians override on their sickest patients, making any headline failing a **0.1 standardised mean difference** balance check unrepresentable
- Strengthened statistical rigour across **6 production pipelines** with propensity matching, B=2000 bootstrap resamples and a Rosenbaum γ plus E-value sensitivity battery, feeding governed dashboards
- Transformed patient event histories into a relapse early-warning model on **Amazon SageMaker**, since relapse, at a 5–15% base rate, shows in event order, not counts
- Gated the model on **6-month** matured outcome labels, then rolled it out to health coaches under per-coach daily alert budgets and an owned alert lifecycle
- Enforced multi-tenant isolation under a Health Insurance Portability and Accountability Act (HIPAA) framework across **10 production pipelines** — tenant-scoped traversal-source wrapper, Parquet partitioning, runtime assertions
- Deployed Datadog observability since nobody knew why the chat platform was unreliable, cutting errors to **under 3%** and revealing ~50 downstream calls per patient message
- Spearheaded the chat service rewrite from Node.js to Go, reaching **100,000 concurrent connections** at P95 under 16 ms in k6-validated burst testing
- Architected a two-stream **Amazon Kinesis** design with Enhanced Fan-Out in on-demand mode, scaling from 4 to **~1,200 shards** under load testing
- Reimagined the data layer across DynamoDB, PostgreSQL and OpenSearch with partition-key and indexing design holding DynamoDB writes under **16 ms P95** at 100,000-user load
- Published React Native apps to 2 app stores with Firebase Cloud Messaging and Apple Push Notification service (APNs), moving **10,000+ patients** off third-party messaging
- Recruited engineering from **0 to 20+** across 6 sub-teams — backend, React Native, web, artificial intelligence, quality assurance (QA), DevOps — standardised on Go and DynamoDB
- Drove a Claude Code and OpenAI Codex rollout across a **20-engineer team**, codifying repository standards in CLAUDE.md and AGENTS.md files to shorten onboarding
- Provisioned Amazon Web Services on Terraform and Elastic Container Service Fargate at **70% Spot** — deploys **under 10 minutes**, infrastructure cost down 25–30%

Lead SDE / Sr. Staff Engineer / Staff Software Engineer

Dec '19 - Feb '25

Deque Software

Hyderabad

- Reframed axe Monitor's bottleneck as deployment, not product: **280 clients**, each on its own instance, version and Keycloak, so every upgrade and trial multiplied
- Implemented multi-tenancy as an organisation identifier on every table, resolving tenant from the JSON Web Token (JWT) into **1 code path**, not database-per-customer
- Steered 70–80% of customers onto shared axe Monitor instances on **Spring Boot 2.x and Java 8/17**, one codebase also serving regulated banks on dedicated databases
- Exposed shared platform application programming interfaces (APIs) to **280 axe Monitor clients** — Jira integration, API versioning, backward compatibility — on one Spring Boot backend
- Championed decomposition of a monolith into **10–15 services**, splitting only where a component needed independent scaling or its own multi-tenant database
- Productized a billing and subscription service for **every Deque product**, restricting features by tier, porting it from Node.js/TypeScript to Java and Spring Boot
- Forged central authentication in Java and Spring Boot on OAuth 2.0, folding every instance's local Keycloak into Red Hat single sign-on with **zero end-user disruption**
- Stewarded every axe Auditor instance from PostgreSQL 9.6 to Amazon Relational Database Service (RDS) PostgreSQL 13.x, then onto Aurora, with **zero downtime**
- Trimmed daily backup windows **3–4 hours** and deploy errors 40% by moving PostgreSQL onto Amazon RDS
- Modernised continuous integration and continuous delivery (CI/CD) from Jenkins, whose jobs stalled at scale, to GitHub Actions, achieving ISO 27001 and **50% faster** release cycles
- Slashed regression testing time **over 50%** with an end-to-end automated testing framework built on Puppeteer
- Tackled accessibility teams' unknown page counts and unannounced changes to uniform resource locators (URLs) with spider discovery **1 to 5 levels deep, or all**
- Orchestrated spider jobs and scheduled axe-core scans for **280 axe Monitor clients**, comparing each run with the last to flag new, fixed and regressed pages
- Constructed comparable test runs in the axe DevTools browser extension, running axe-core on **every page and user action** in a flow, de-duplicating issues server-side
- Broadened axe DevTools scanning into cross-origin iframes, which the browser's security model hid and where the extension reported **zero issues**, landing phase 1
- Authored the **axe Model Context Protocol (MCP) server** in 2024, giving AI agents axe accessibility testing, and rolled it out to **every enterprise customer**

Software Engineer III

Aug '18 - Jul '19

Rocket Software

Bangalore

- Anchored decomposition of the **IBM Db2 Configuration Manager** monolith into Java and Spring Boot microservices, defining boundaries across z/OS mainframe and distributed platforms
- Surfaced defect hot-spots by isolating services behind representational state transfer (REST) interfaces, reducing reported bugs **over 50%**
- Partnered with product to extend platform support to Windows and three Linux distributions — Ubuntu, CentOS, Red Hat Enterprise Linux (RHEL) — raising sales **over 20%**
- Coached a **5-engineer** team through delivery planning, code review and performance appraisals, remaining hands-on as an individual-contributor Software Engineer III

Co-Founder & VP of Technology

Mar '17 - Apr '18

VoltusWave Technologies

Hyderabad

- Co-founded VoltusWave and released its first no-code application Platform-as-a-Service (aPaaS) for web and Android on Amazon Elastic Container Service, winning **8 early enterprise clients**

- Mobilized an engineering organisation from **0 to 35** in 13 months, mentoring freshers and seniors and setting the architectural roadmap later products followed

Senior Software Engineer

Mar '16 - Mar '17

Teletext India

Hyderabad

- Codified a MuleSoft enterprise service bus scatter-gather giving sales reps **~30 suppliers' rates** in parallel, superseding .NET 2.0 code a prior team could not upgrade
- Retired a third-party hotel pricing search whose load-based billing grew with **2–3 million daily requests** at peak, calling suppliers directly instead
- Cached airport-destination-date queries on in-house Elasticsearch, Logstash and Kibana (ELK) with a 20-minute time-to-live, suppliers called only on a miss, lowering operational cost **46%**
- Traced hidden hotel inventory to clashing identifiers across ~30 suppliers, where unmapped properties lacked images and ratings, and adopted licensed **GIATA MultiCodes**
- Mechanised the reconciliation pipeline on Amazon Web Services (AWS) Lambda with human approval retained, growing revenue **8% month over month**
- Lifted hotel booking conversions **65%** by widening matched inventory, so each hotel's lowest supplier rate could be shown, at 2–3 million daily requests

Technical Architect

May '15 - Nov '15

CURA Software Solutions

Hyderabad

- Pinpointed **2 paid Microsoft licences**, Windows Server and SQL Server, as the biggest customer cost objection to a Governance, Risk and Compliance (GRC) platform
- Recast the SQL Server data model as MySQL on Ubuntu and Node.js, a **2nd supported stack** for new enterprise accounts only, leaving Windows customers untouched
- Unlocked **12 new enterprise customers** in two months, October and November 2015, on the licence-free Linux deployment

Technical Lead

Jul '13 - May '15

innRoad India Hotel Software

Hyderabad

- Replatformed a server-rendered ASP.NET application into an AngularJS single-page application (SPA) over .NET Windows Communication Foundation (WCF) services, **splitting 1 app into 2 tiers**
- Calibrated the data layer on **3 fronts**, stored procedures, table design and indexing, clearing the query bottlenecks behind slow screen loads
- Shortened screen load times **60–70%** once the single-page client and tuned queries replaced server-rendered pages

Software Architect

Jun '12 - May '13

McDonald's Corporation

Oak Brook, IL

- Unified employee records from every store for leadership in a **2-country** Master Data Management (MDM) pilot, United States and Canada, on SQL Server
- Salvaged the unusable built-in MDM interface with a rebuilt front end, adding **first seen and last updated** timestamps per field on every master record

Senior Java Developer

Apr '08 - May '12

El Paso Corporation

Houston, TX

- Translated a Citrix-hosted Java application into JavaServer Pages (JSP) and Java Enterprise Edition for Tennessee Gas Pipeline, an **11,800-mile** interstate system
- Evolved **3 core modules** — Nominations, Flowing Gas and Contracts — as requirements arrived, covering scheduling and contracting for interstate gas transport over 4 years
- Refactored the scheduling modules' schema and stored procedures through a **SQL Server 2000 to 2005** migration, owning table and index design and query tuning

.NET Developer

Apr '07 - Apr '08

LyntonWeb

Houston, TX

- Bridged suppliers and buyers through a .NET portal on SQL Server 2005, integrating **Mexican PayPal** to settle payments inside the portal
- Seeded portal adoption with a Windows service sweeping **every invoice** from a watched folder to the server, built to move everyone onto the portal

EDUCATION

MS Computer Science

Aug '04 - Dec '07

University of Houston

PG Diploma in Artificial Intelligence / Machine Learning

Jan '21

IIIT Bangalore